



CS7.404: Digital Image Processing

Monsoon 2025: Image Restoration



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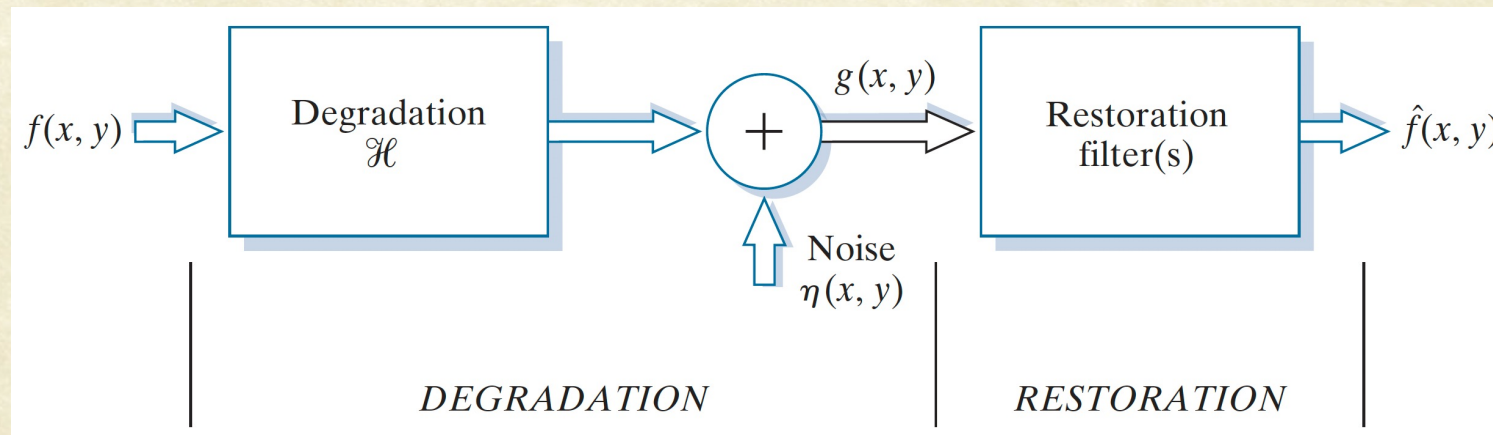
A Quick Recap



A Model for Image Degradation and Restoration

- Image Restoration

- Use a priori knowledge of the degradation
- Modeling the degradation and apply the inverse process
- Formulate and evaluate objective criteria of goodness



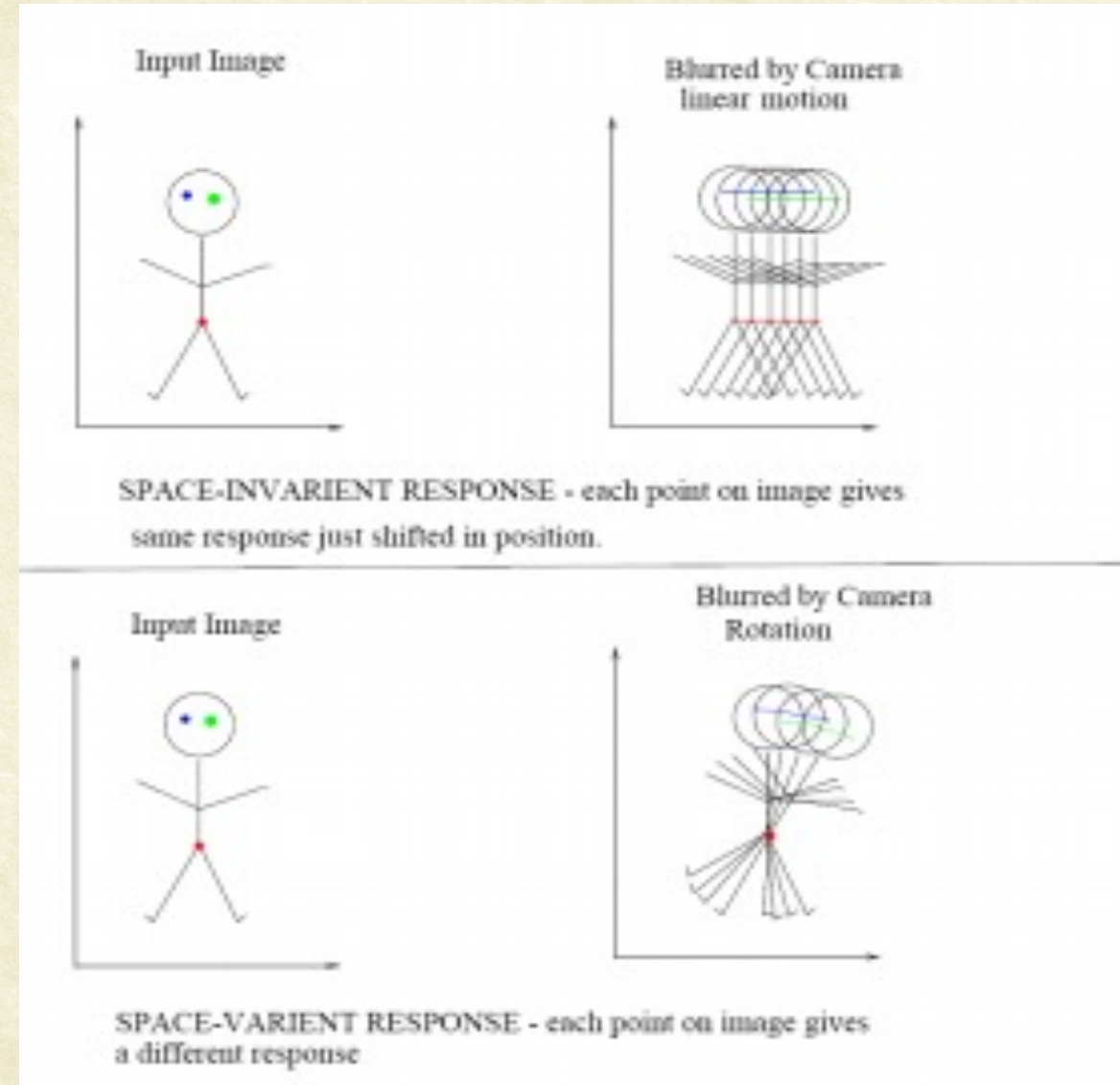
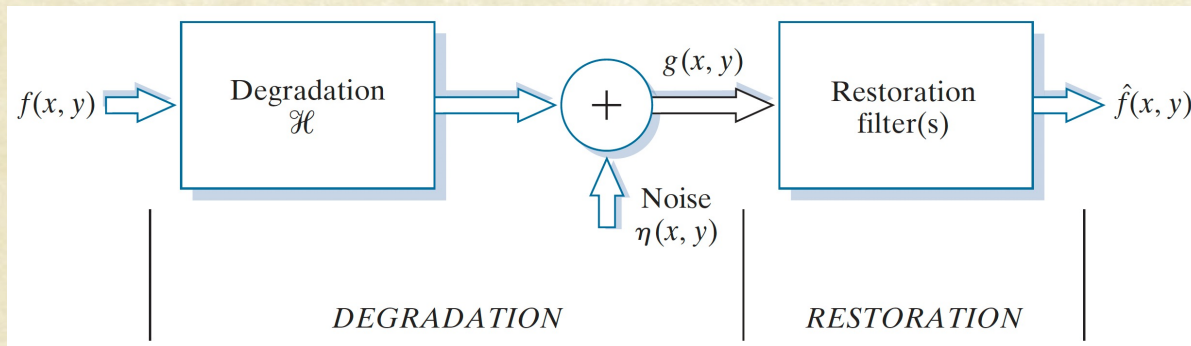
$$g(x, y) = H[f(x, y)] + \eta(x, y)$$

- Design the restoration filters such that $\hat{f}(x, y)$ is as close to $f(x, y)$ as possible.



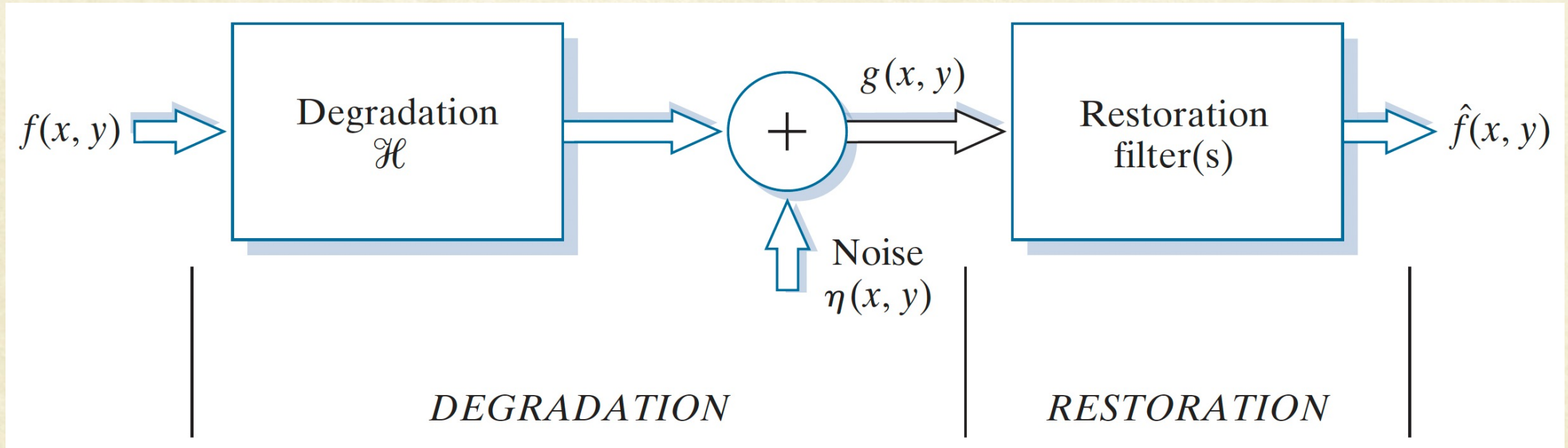
Assumptions for the Distortion Model

- Noise
 - Independent of spatial location
 - except for periodic noise
 - Uncorrelated with the image
- Degradation function, H
 - Linear
 - Position-invariant





Mathematical model of Image Degradation/Restoration



$$g(x, y) = H[f(x, y)] + \eta(x, y)$$

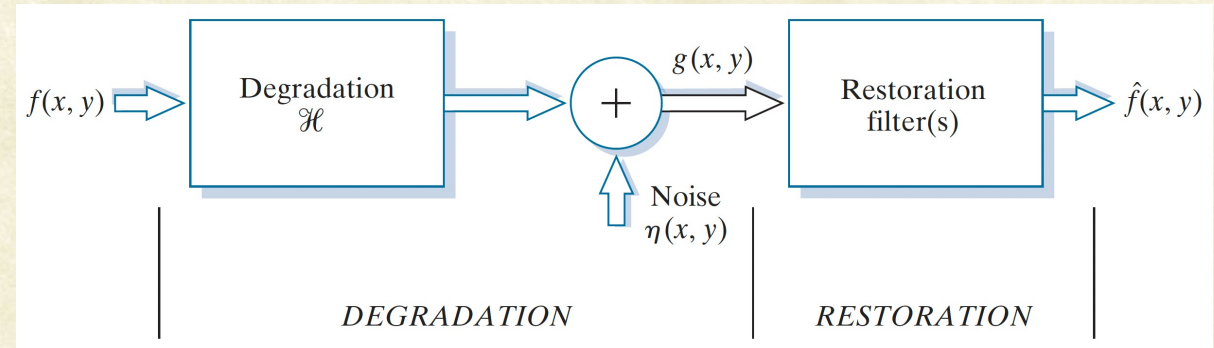
If H is a linear, position-invariant process,

$$g(x, y) = h(x, y) \star f(x, y) + \eta(x, y), \text{ or}$$
$$G(u, v) = H(u, v) F(u, v) + N(u, v)$$



Divide and Conquer: Step #1

- Image degraded only by noise



- Assuming 'H' is identity, the model reduces to:

$$g(x, y) = f(x, y) + \eta(x, y)$$

or

$$G(u, v) = F(u, v) + N(u, v)$$



Noise Models

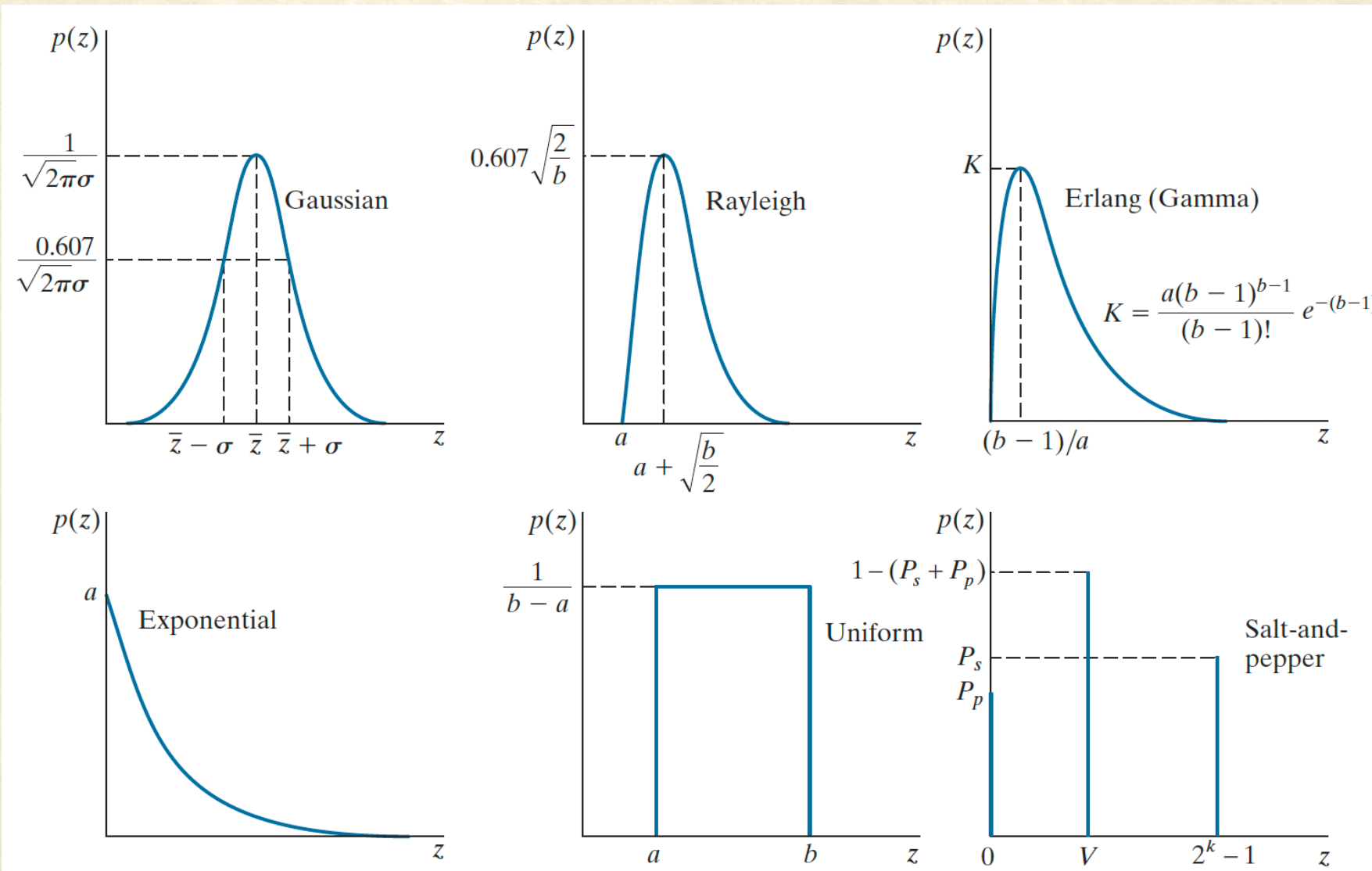
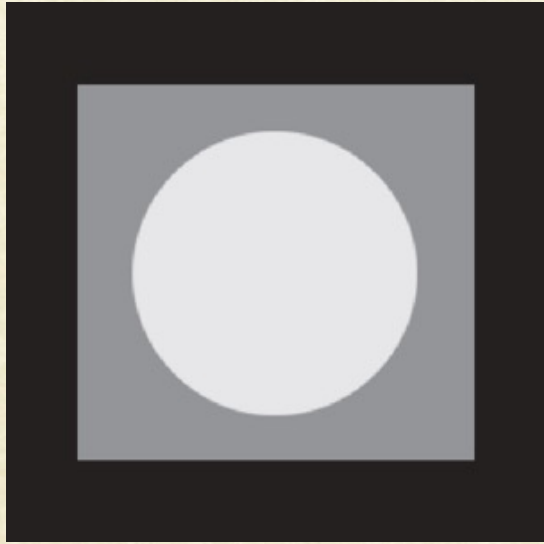
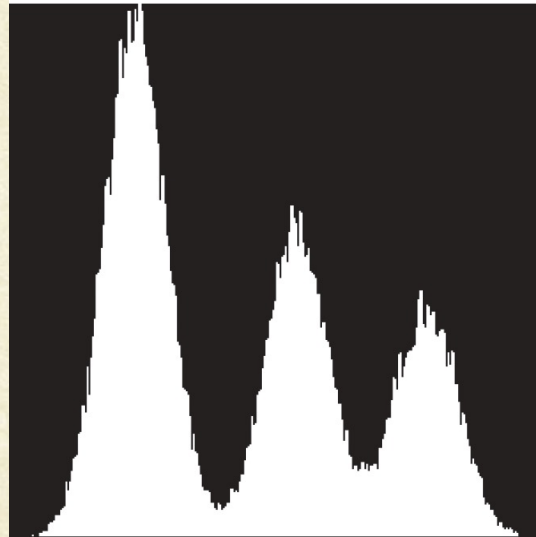
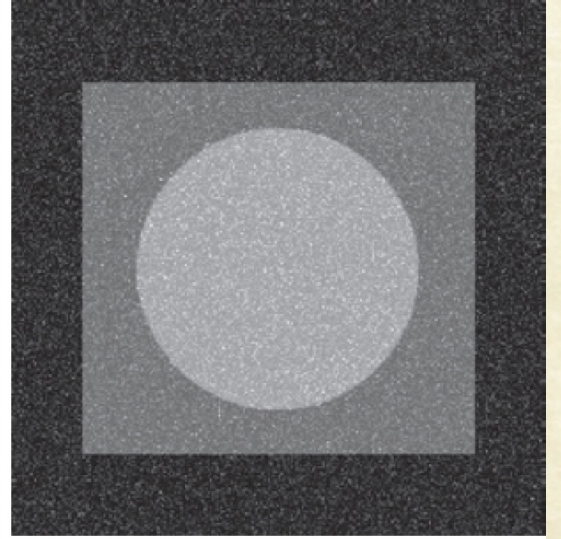
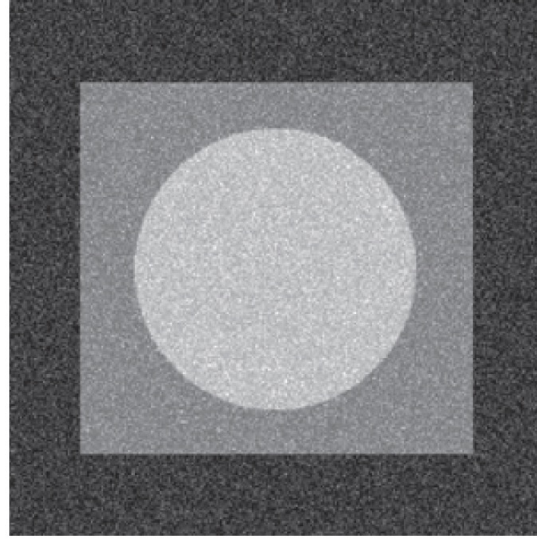
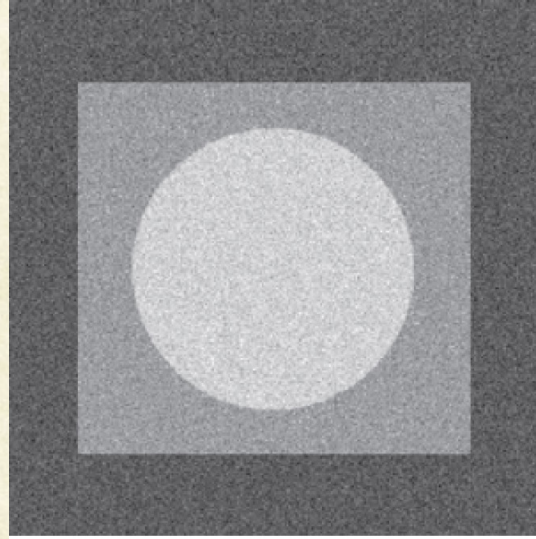




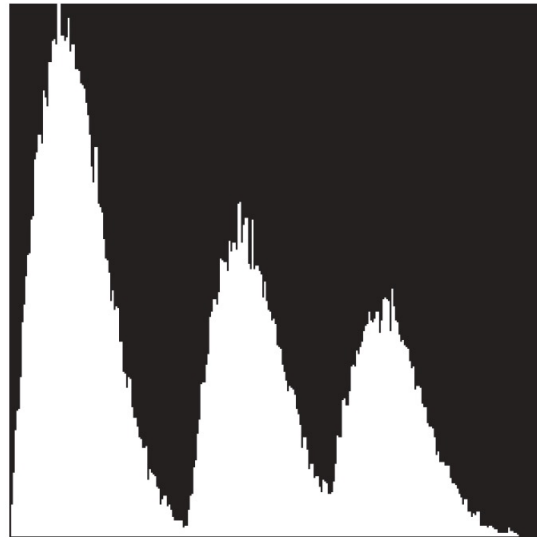
Illustration of Noise Models



Original Image



Gaussian



Rayleigh

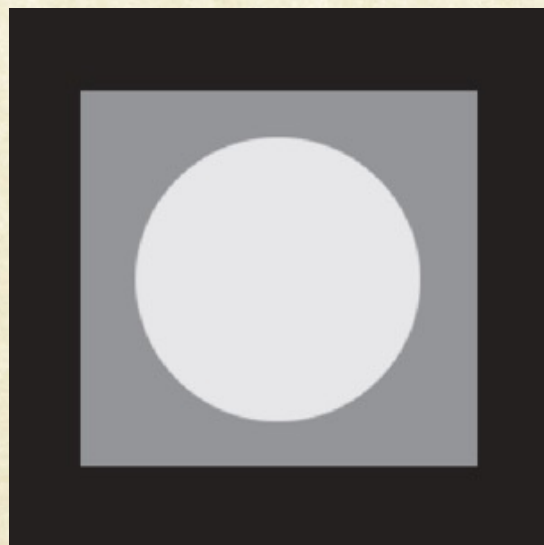


Erlang

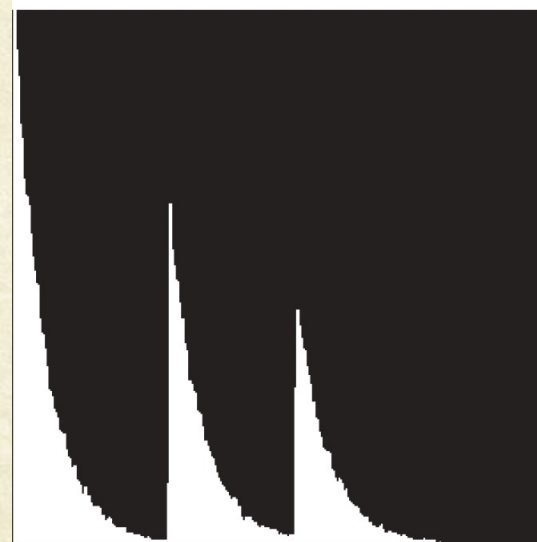
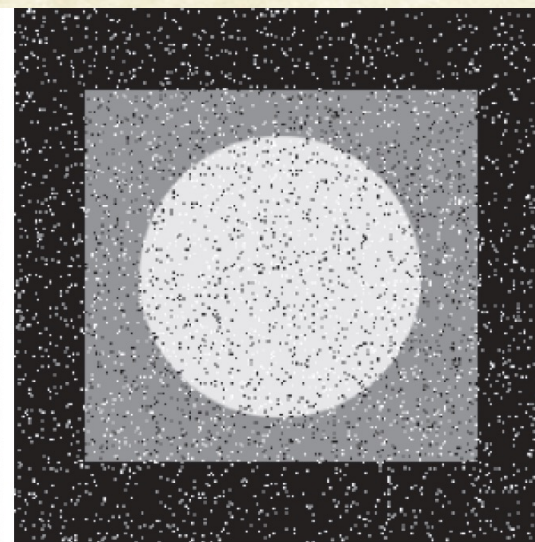
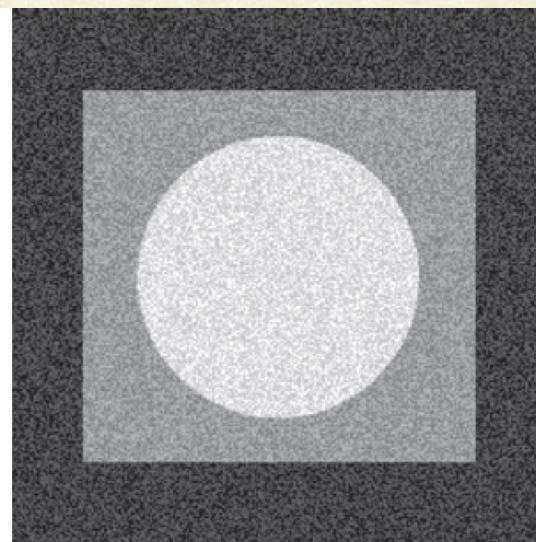
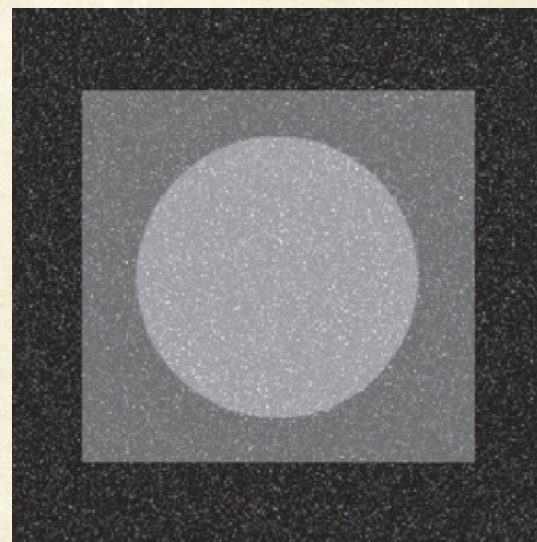
- Visually similar.
- Not easy to determine noise model from appearance



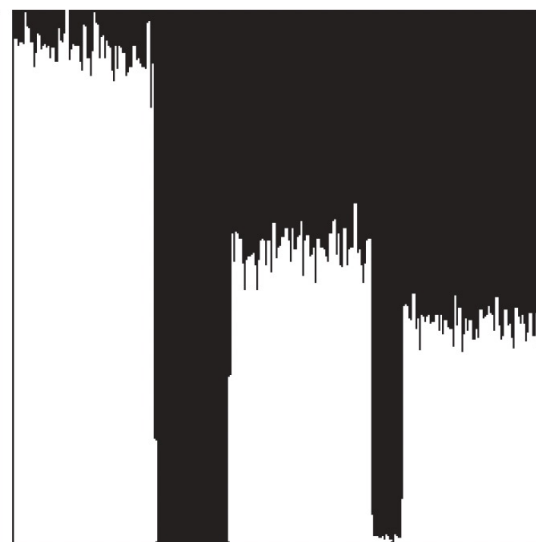
Illustration of Noise Models



Original Image



Exponential



Uniform



Salt & Pepper

- Visually similar.
- Not easy to determine noise model from appearance



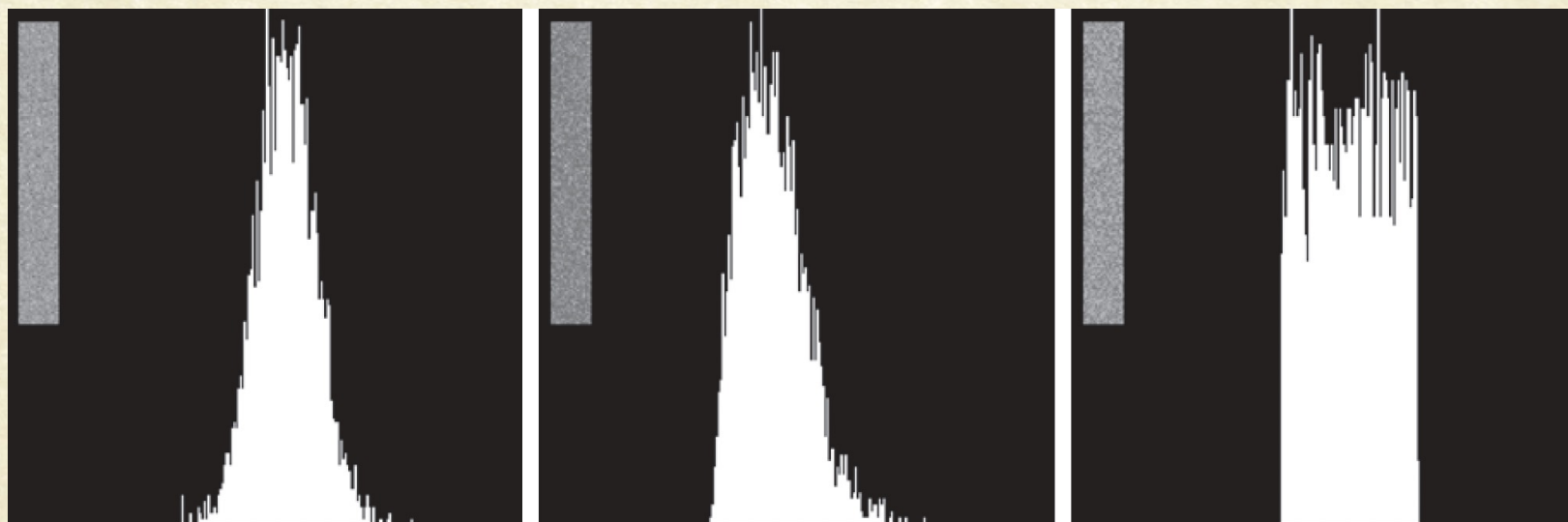
Understanding System Noise

- Case 1: Imaging system available
 - Noise Calibration: Capture a set of 'flat environments' (e.g., solid gray board, object at fixed location)
 - Select the model with better statistical test scores (Goodness of fit measures like Akaike Information Criteria (AIC) or Likelihood Ratio Test (LRT))
 - Compute model parameters (mean, variance, etc.) from the statistics of pixel values.



Understanding System Noise

- Case 2: Only images available
 - Estimate from patches of constant intensity
 - For impulse noise, use a mid-gray patch/area



Gaussian

Rayleigh

Uniform



Restoration (in presence of noise only)

- Mean filters

- Arithmetic mean filter

$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(r, c) \in S_{xy}} g(r, c)$$

- Geometric mean filter

$$\hat{f}(x, y) = \left[\prod_{(r, c) \in S_{xy}} g(r, c) \right]^{\frac{1}{mn}}$$





Restoration (in presence of noise only)

- Mean filters
 - Harmonic mean filter (Gaussian, Salt)

$$\hat{f}(x, y) = \frac{mn}{\sum_{(r,c) \in S_{xy}} \frac{1}{g(r, c)}}$$

- Contraharmonic mean filter (Salt & Pepper)

$$\hat{f}(x, y) = \frac{\sum_{(r,c) \in S_{xy}} g(r, c)^{Q+1}}{\sum_{(r,c) \in S_{xy}} g(r, c)^Q}$$

Q : order of the filter; Good for salt and pepper noise

Eliminates pepper noise for $Q > 0$ and salt noise for $Q < 0$

NB: cf. arithmetic filter if $Q = 0$, harmonic mean if $Q = -1$



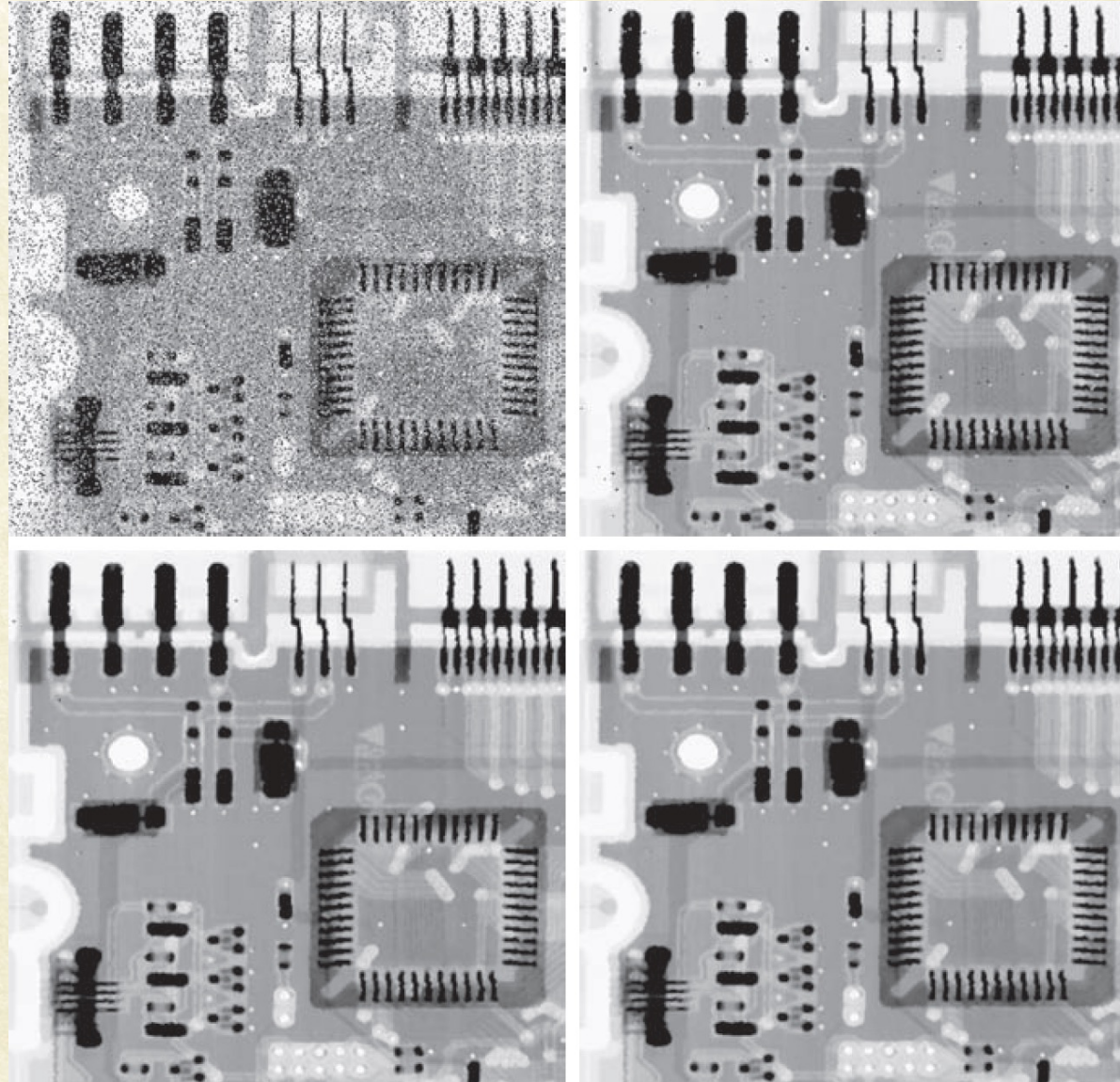
Restoration (in presence of noise only)

- Median filter

a	b
c	d

FIGURE 5.10

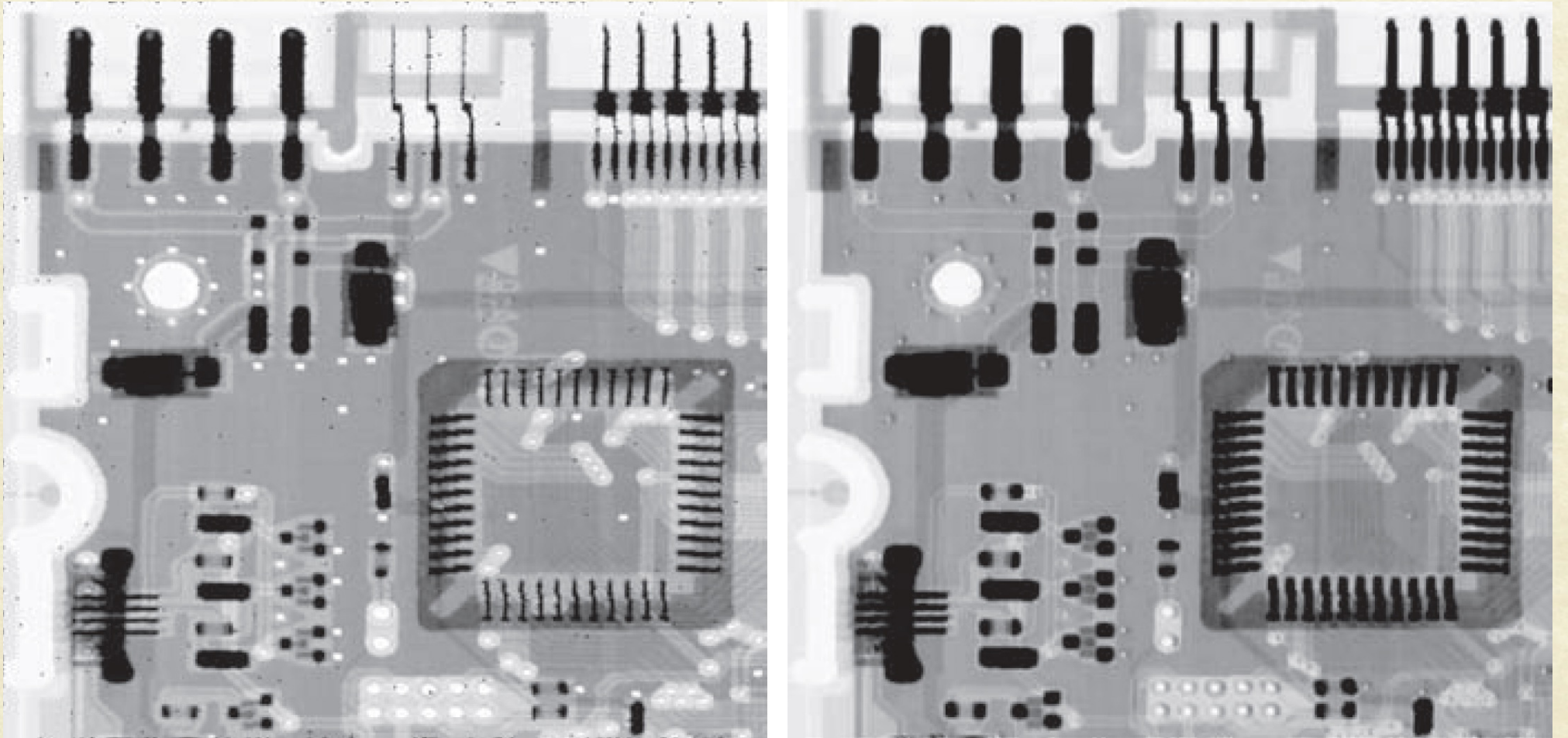
(a) Image corrupted by salt-and-pepper noise with probabilities $P_s = P_p = 0.1$. (b) Result of one pass with a median filter of size 3×3 . (c) Result of processing (b) with this filter. (d) Result of processing (c) with the same filter.





Restoration (in presence of noise only)

- Max, Min filters



a b

FIGURE 5.11

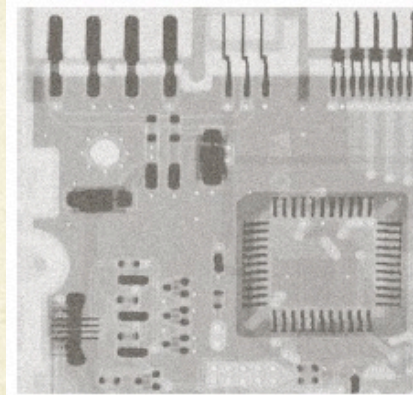
(a) Result of filtering Fig. 5.8(a) with a max filter of size 3×3 .

(b) Result of filtering Fig. 5.8(b) with a min filter of the same size.

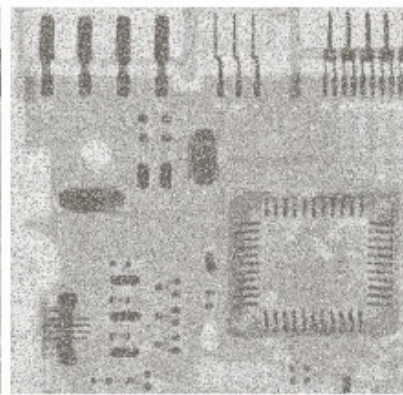


Restoration (in presence of noise only)

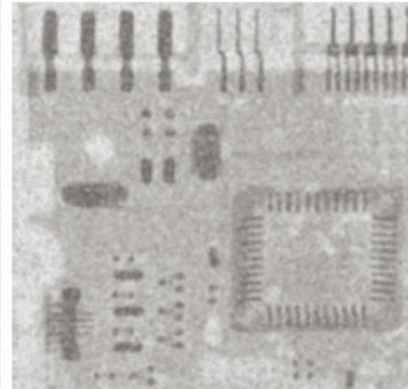
original



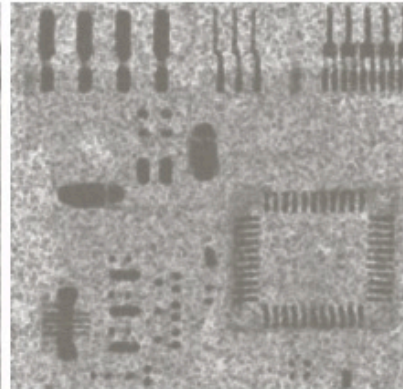
Original + salt and pepper noise



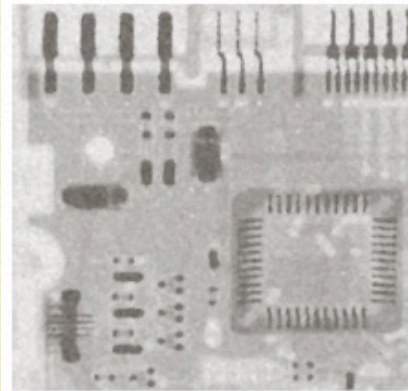
Arithmetic mean filter



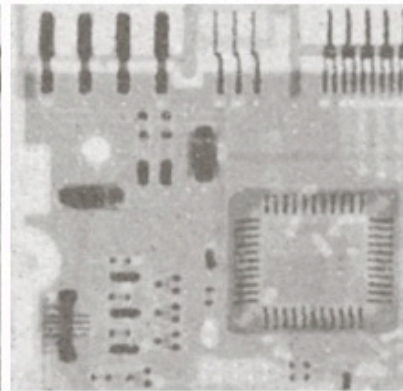
Geometric mean filter



Median filter



Alpha Trimmed filter

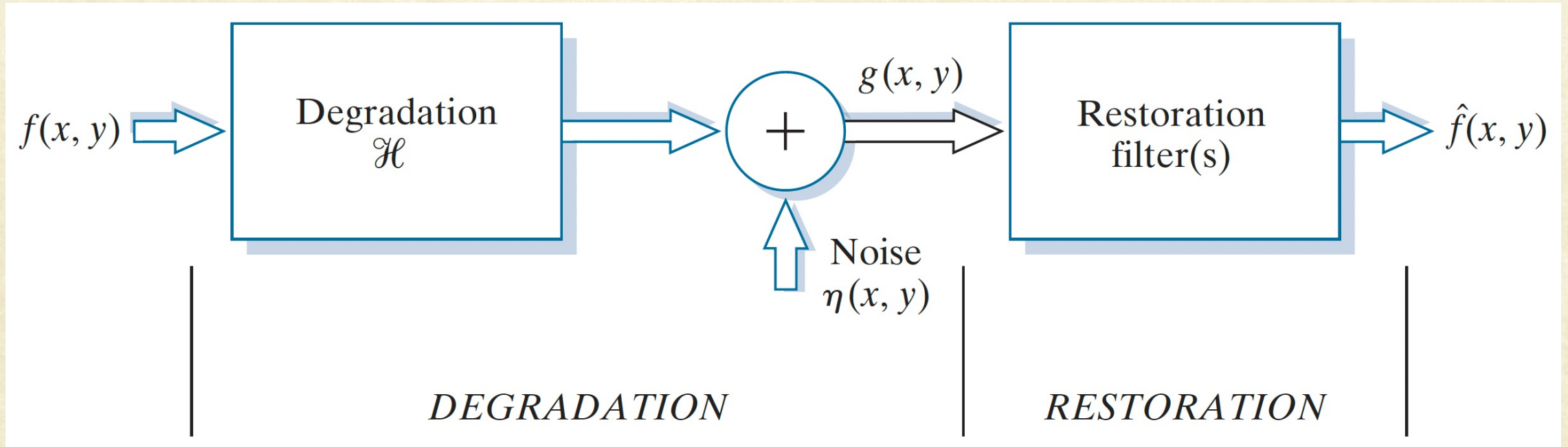




Questions?



Image Restoration: Periodic Noise



$$g(x, y) = h(x, y) \star f(x, y) + \eta(x, y), \text{ or}$$
$$G(u, v) = H(u, v) F(u, v) + N(u, v)$$

Assuming only additive noise: $g(x, y) = f(x, y) + \eta(x, y)$



Adaptive Local Noise Reduction Filter

- Mean filter blurs edges while removing additive noise
- Adaptive filter: Reduce averaging around edges

$$\hat{f}(x, y) = g(x, y) - \frac{\sigma_{\eta}^2}{\sigma_L^2} [g(x, y) - m_L]$$

- $\sigma_{\eta}^2 = 0$: Image is noise free
- $\sigma_L^2 \gg \sigma_{\eta}^2$: Local variation is very high (presence of edges)
- $\sigma_L^2 = \sigma_{\eta}^2$: Local variation similar to overall noise (mean filter)



Adaptive mean filtering

a	b
c	d

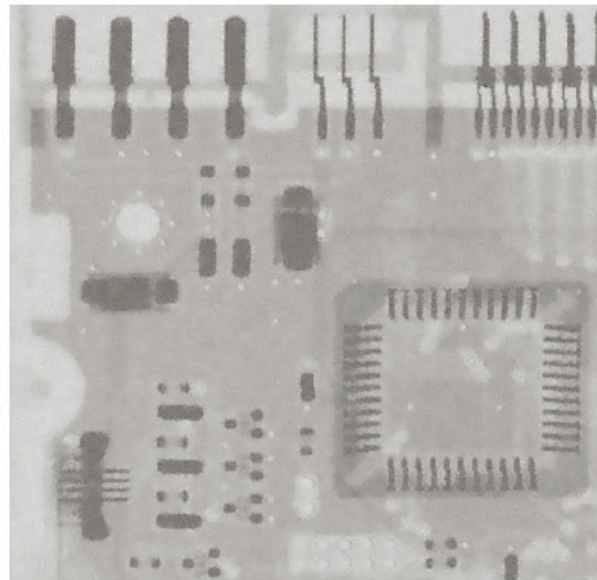
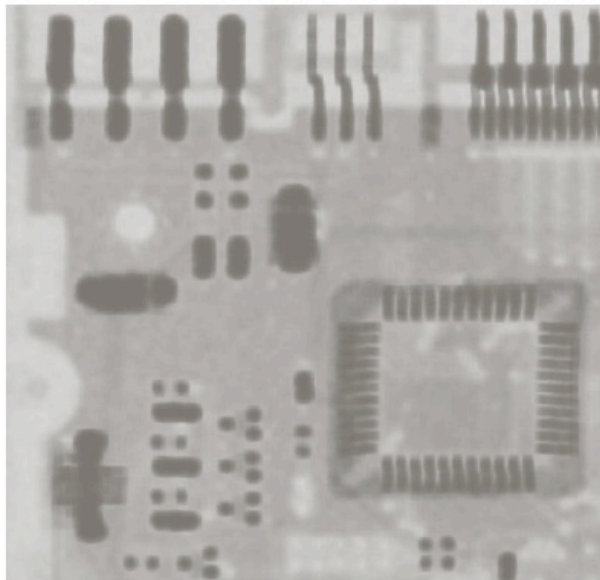
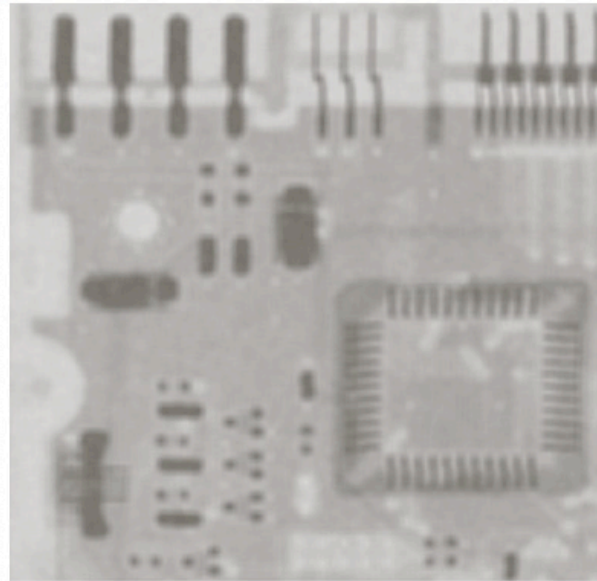
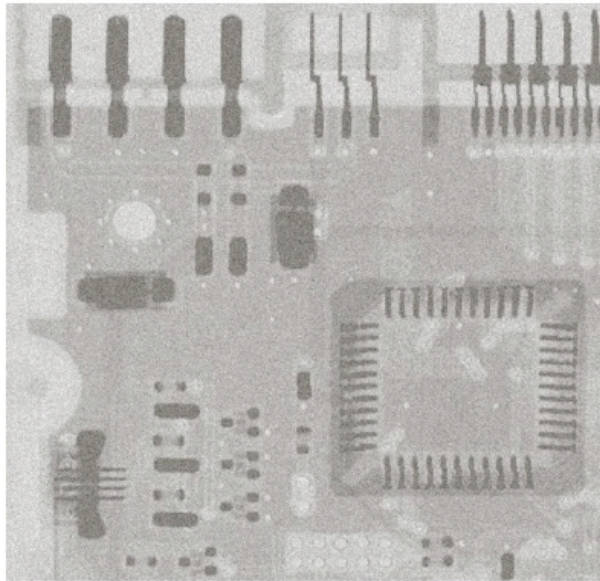
FIGURE 5.13

(a) Image corrupted by additive Gaussian noise of zero mean and variance 1000.

(b) Result of arithmetic mean filtering.

(c) Result of geometric mean filtering.

(d) Result of adaptive noise reduction filtering. All filters were of size 7×7 .

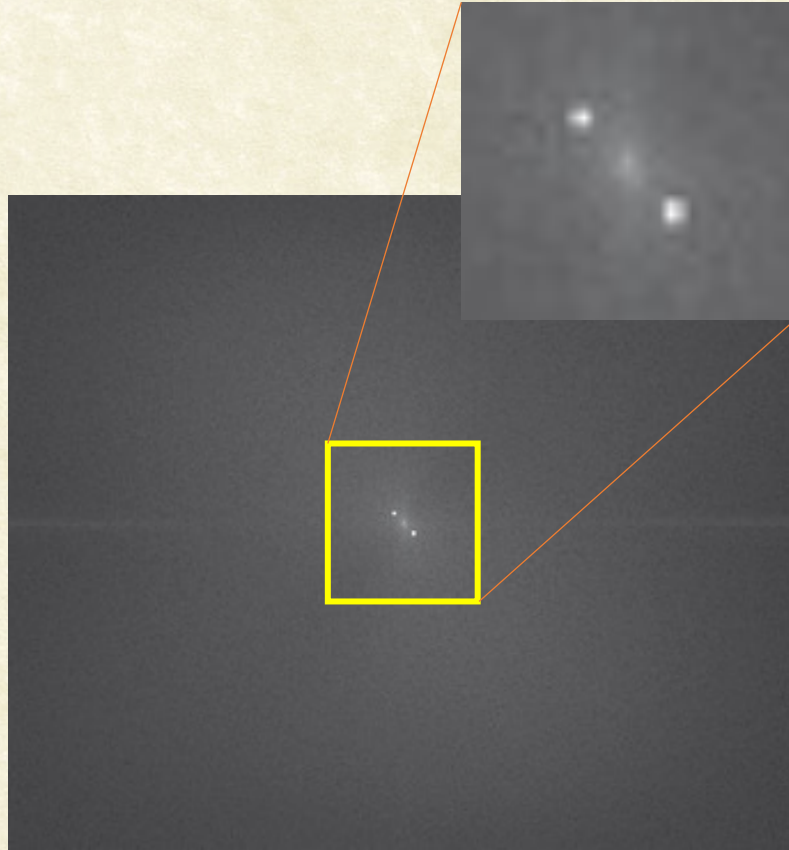




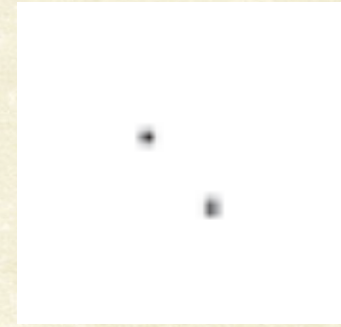
Additive Sinusoidal Noise



Additive noise of regular frequencies are easier to handle in the frequency domain.



x

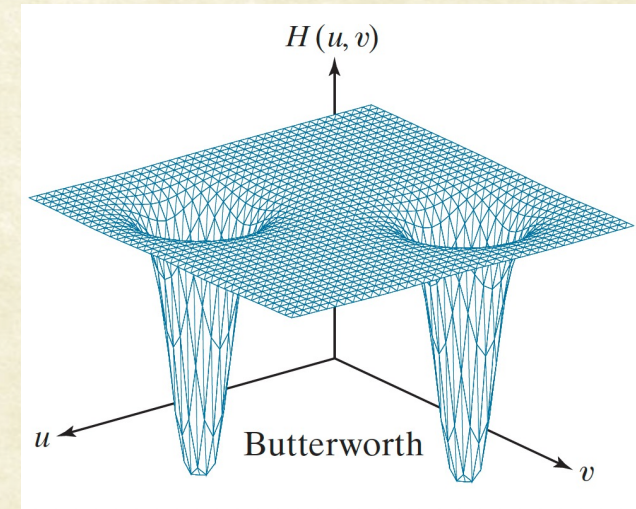
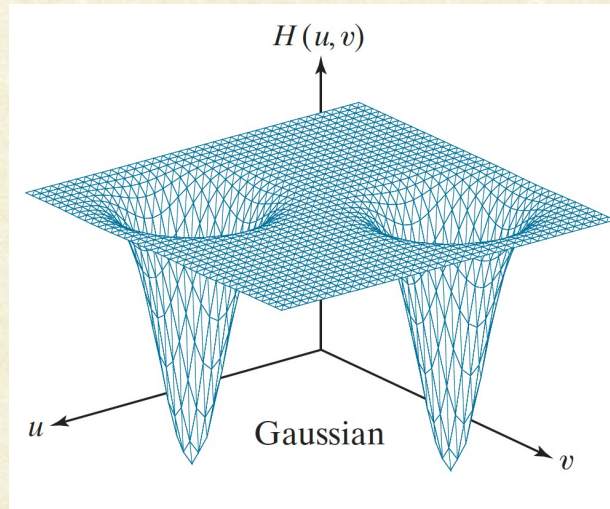
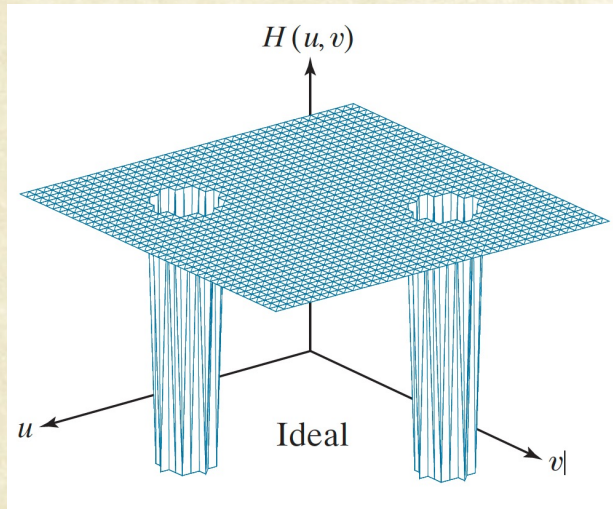


Very difficult to get result of this quality via spatial domain filtering



Possible Notch Filters

- $H_{NR}(u, v) = \prod_{k=1}^Q H_k(u, v) H_{-k}(u, v)$

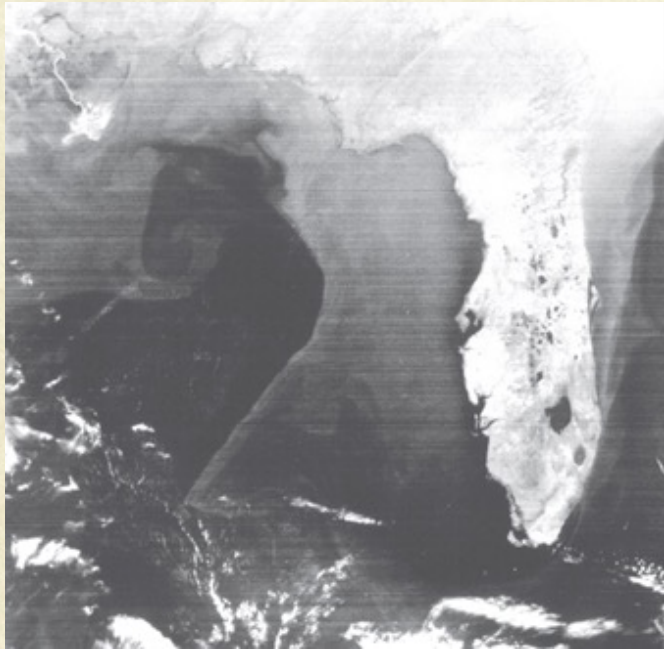


- $H_{NP}(u, v) = 1 - H_{NR}(u, v)$

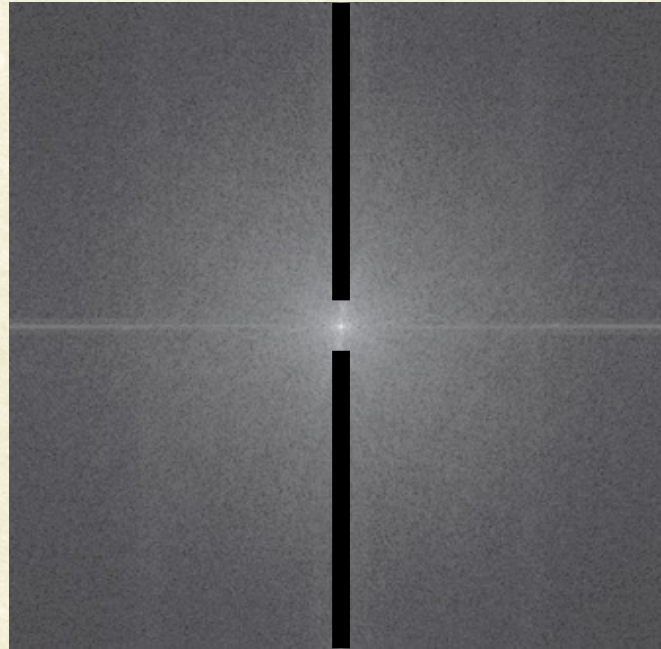


Restoration (in presence of noise only)

- Notch pass/reject



Satellite image of Florida and the Gulf of Mexico. (Note horizontal sensor scan lines.)



Spectrum of image and Notch reject filter transfer function.



Filtered image. (Original image courtesy of NOAA.)



Optimum Notch Filtering

- Qn: What is the periodic noise is more complex? Cannot suppress the noise without losing significant image signal.
- Solution:
 - Estimate the noise using notch-pass filter

$$\eta(x, y) = \mathfrak{S}^{-1} \{ H_{\text{NP}}(u, v) G(u, v) \}$$

- Subtract a weighted portion of the noise image from degraded image. Goal: Minimize the local variance of the image

$$\hat{f}(x, y) = g(x, y) - w(x, y)\eta(x, y)$$

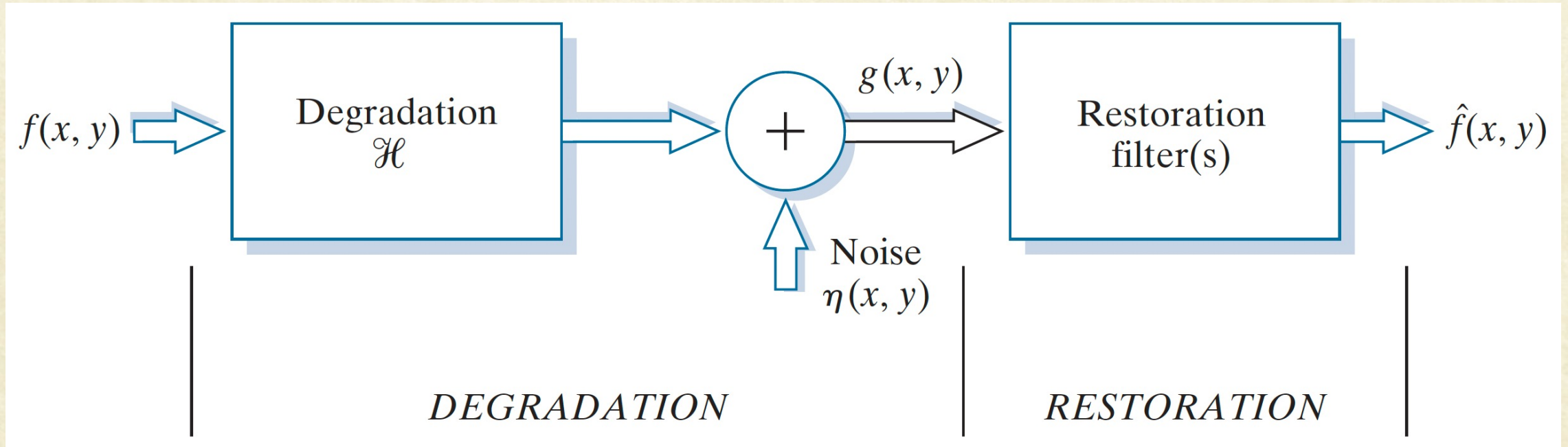
$$w(x, y) = \frac{\overline{g\eta} - \bar{g}\bar{\eta}}{\overline{\eta^2} - \bar{\eta}^2}$$



Questions?



Image Restoration: Modeling Degradation, $H(u, v)$



$$g(x, y) = h(x, y) \star f(x, y) + \eta(x, y), \text{ or}$$

$$G(u, v) = H(u, v) F(u, v) + N(u, v)$$

Assuming no noise, $G(u, v) = H(u, v) F(u, v)$

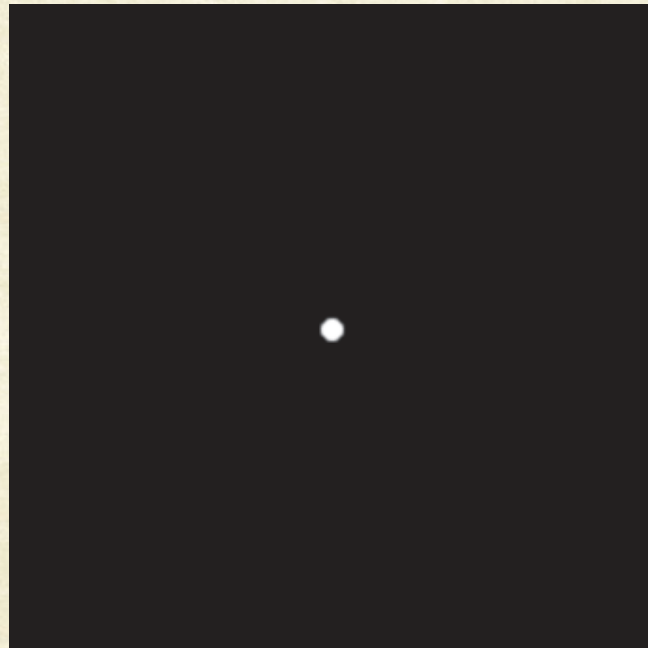


Estimation of degradation function

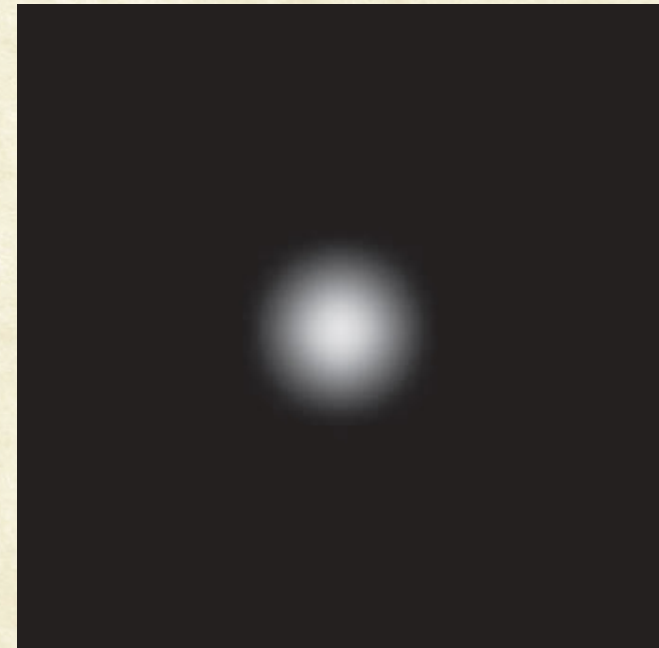
- Three main ways:
 - Observation → look, find, iterate
 - Experimentation → If you have access to the imaging system
 - Mathematical modelling

$$H(u, v) = \frac{G(u, v)}{A}$$

Estimating a degradation by impulse characterization.



An impulse of light (magnified).



Imaged (degraded) impulse.



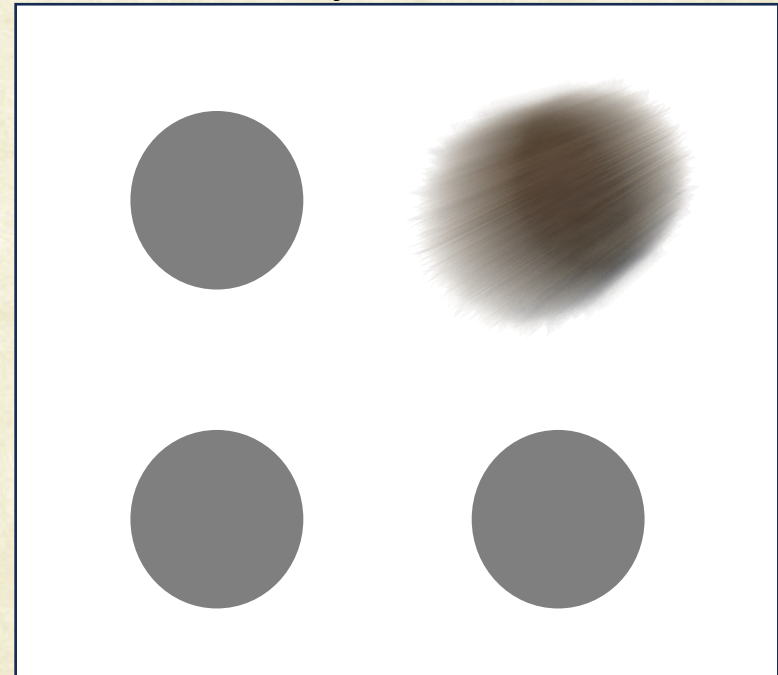
Motion Blur

- During Long Exposure, if amount of light hitting the sensor changes significantly → Motion Blur
- Causes (one or more of):

Camera motion



Subject motion





Motion blur effect

```
#define filterWidth 9
#define filterHeight 9

double filter[filterHeight][filterWidth] =
{
    1, 0, 0, 0, 0, 0, 0, 0, 0,
    0, 1, 0, 0, 0, 0, 0, 0, 0,
    0, 0, 1, 0, 0, 0, 0, 0, 0,
    0, 0, 0, 1, 0, 0, 0, 0, 0,
    0, 0, 0, 0, 1, 0, 0, 0, 0,
    0, 0, 0, 0, 0, 1, 0, 0, 0,
    0, 0, 0, 0, 0, 0, 1, 0, 0,
    0, 0, 0, 0, 0, 0, 0, 1, 0,
    0, 0, 0, 0, 0, 0, 0, 0, 1,
};

double factor = 1.0 / 9.0;
double bias = 0.0;
```





Estimation by Modeling: Motion Blur

- Assume planar motion in the direction: $(x_0(t), y_0(t))$

$$g(x, y) = \int_0^T f[x - x_0(t), y - y_0(t)] dt$$

- FT of degraded image:

$$G(u, v) = F(u, v) \int_0^T e^{-j2\pi[ux_0(t)+vy_0(t)]} dt$$

- We know that $G(u, v) = H(u, v) F(u, v)$. Hence,

$$H(u, v) = \int_0^T e^{-j2\pi[ux_0(t)+vy_0(t)]} dt$$





Estimation by Modeling: Uniform Motion Blur

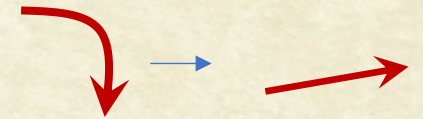
Degradation Kernel:

$$H(u, v) = \int_0^T e^{-j2\pi[ux_0(t)+vy_0(t)]} dt$$

Uniform Motion:

$$x_0(t) = at/T \quad \text{and} \quad y_0(t) = bt/T$$

$$H(u, v) = \frac{T}{\pi(ua + vb)} \sin[\pi(ua + vb)] e^{-j\pi(ua + vb)}$$





Estimation by Modeling (atmospheric turbulence)



Degradation model proposed by Hufnagel and Stanley [1964] based on the physical characteristics of atmospheric turbulence:

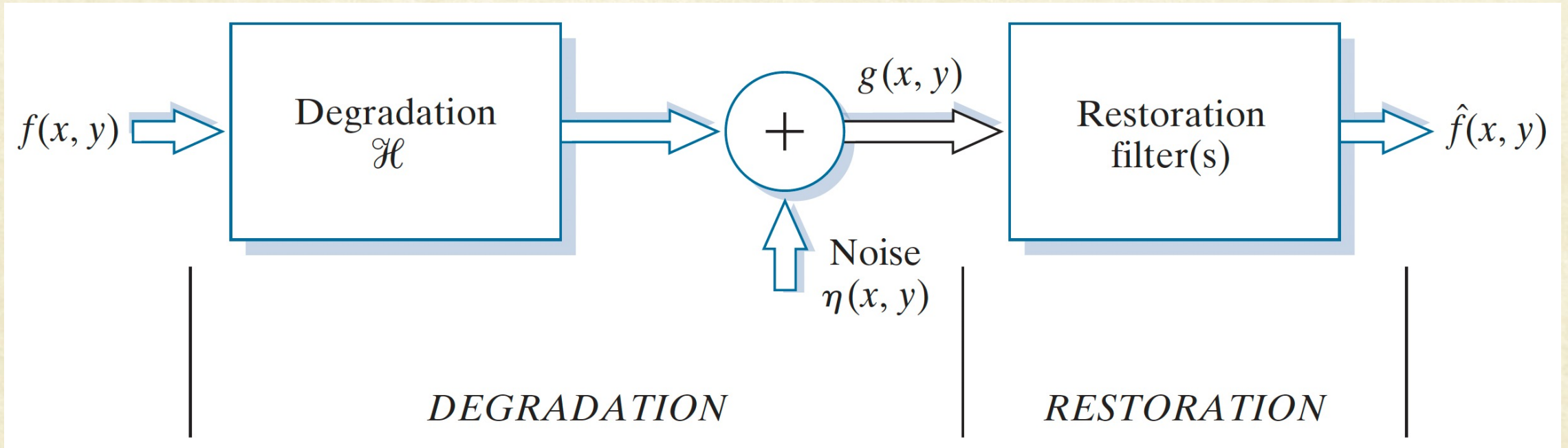
$$H(u, v) = e^{-k(u^2 + v^2)^{5/6}}$$



No Turbulence $k = 0$	Severe Turbulence $k = 0.0025$
Mild Turbulence $k = 0.001$	Low Turbulence $k = 0.00025$



Image Restoration: Handling $H(u, v)$ and $N(u, v)$



$$g(x, y) = h(x, y) \star f(x, y) + \eta(x, y), \text{ or}$$
$$G(u, v) = H(u, v) F(u, v) + N(u, v)$$



Recovering Image (both Noise and degradation)

- Even if we know the $H(u,v)$, we cannot recover the original image!!

$$- \hat{F}(u, v) = \frac{G(u, v)}{H(u, v)}$$

$$- G(u, v) = H(u, v)F(u, v) + N(u, v)$$

$$- \hat{F}(u, v) = F(u, v) + \frac{N(u, v)}{H(u, v)}$$

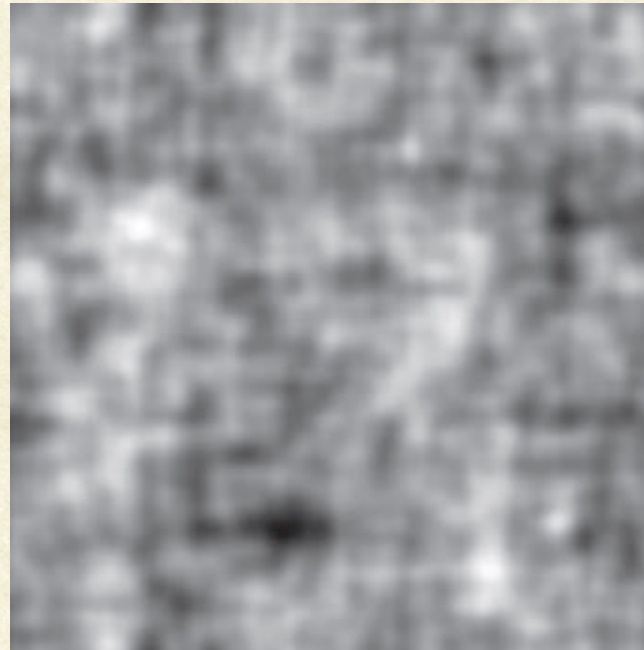
- Two problems:
 1. $N(u,v)$ is a random function whose Fourier Transform is not known
 2. If degradation has zero or small values $\rightarrow N(u,v)/H(u,v)$ will dominate



$H(u,v) + \text{Noise}$



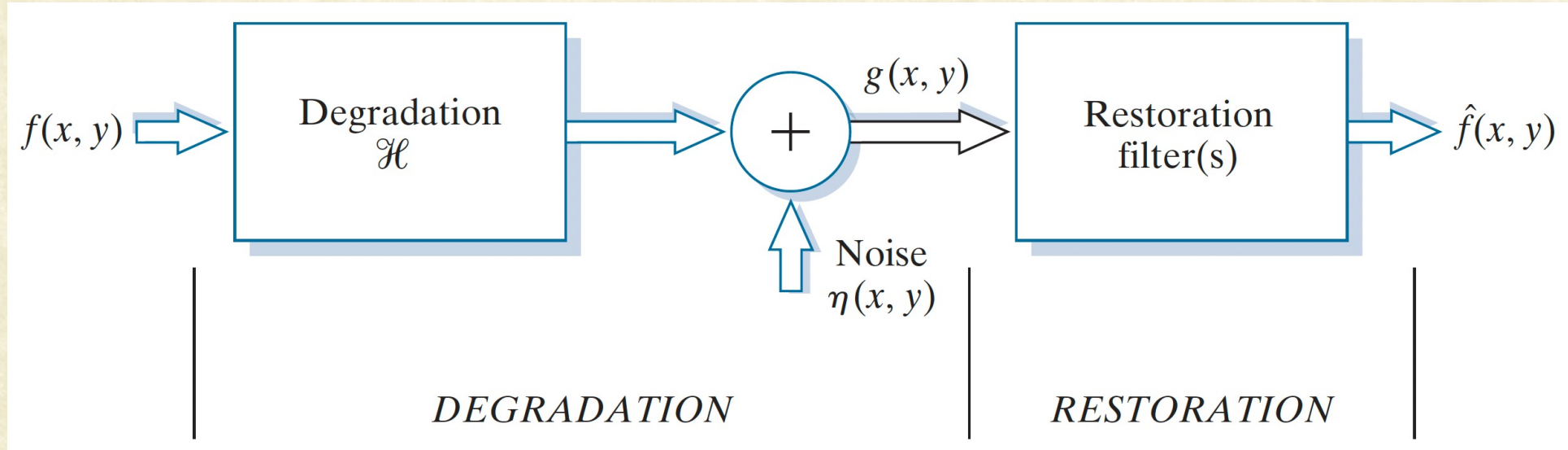
Degraded Image
(with known model)



Full Filter	H Cutoff > 85
H Cutoff > 70	H Cutoff > 40



Wiener filter



Consider image and noise as random variables,
Mean Squared Error of restoration:

$$e^2 = E \left\{ (f - \hat{f})^2 \right\}$$

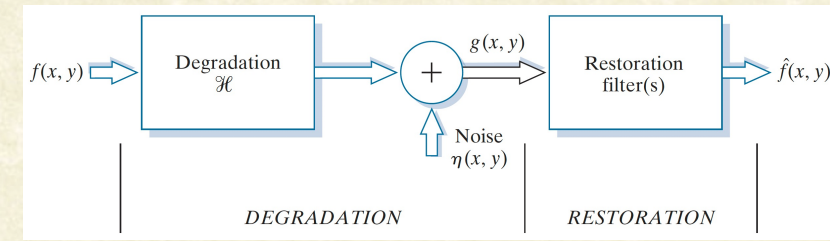
Assumption:

- Noise and image are uncorrelated



Wiener filter

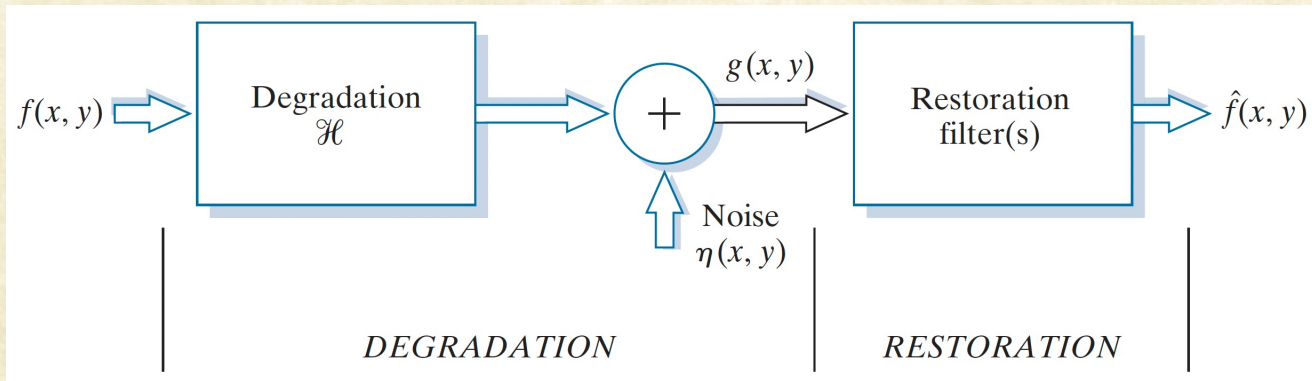
$$e^2 = E \left\{ (f - \hat{f})^2 \right\}$$



$$\begin{aligned}\hat{F}(u, v) &= \left[\frac{H^*(u, v) S_f(u, v)}{S_f(u, v) |H(u, v)|^2 + S_\eta(u, v)} \right] G(u, v) \\ &= \left[\frac{H^*(u, v)}{|H(u, v)|^2 + S_\eta(u, v) / S_f(u, v)} \right] G(u, v) \\ &= \left[\frac{1}{H(u, v)} \frac{|H(u, v)|^2}{|H(u, v)|^2 + S_\eta(u, v) / S_f(u, v)} \right] G(u, v)\end{aligned}$$



Wiener filter



$$e^2 = E \left\{ (f - \hat{f})^2 \right\}$$

The minimum of the error function e is given by:

$$\hat{F}(u, v) = \left[\frac{1}{H(u, v)} \frac{|H(u, v)|^2}{|H(u, v)|^2 + S_\eta(u, v)/S_f(u, v)} \right] G(u, v)$$

$S_\eta(u, v) = |N(u, v)|^2 =$ Power spectrum of the noise (autocorrelation of noise)

$S_f(u, v) = |F(u, v)|^2 =$ Power spectrum of the undegraded image



Wiener filter

- When two spectrums are not known or cannot be estimated, the equation is approximated as:

$$\hat{F}(u, v) = \left[\frac{1}{H(u, v)} \frac{|H(u, v)|^2}{|H(u, v)|^2 + K} \right] G(u, v)$$



Wiener Filter Results



Full Filter



Radially Limited Filter



Wiener Filter



Wiener Filter for Motion Blur + Noise

Image with motion blur and additive
noise, restored using inverse filtering and
Wiener filter

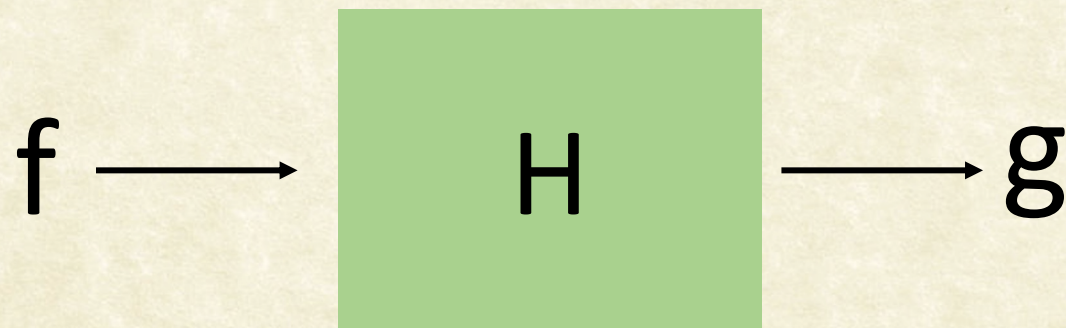
Same sequence with an order of
magnitude less noise variance

Same sequence with five orders of
magnitude less noise variance





Image Restoration



Inverse
problems



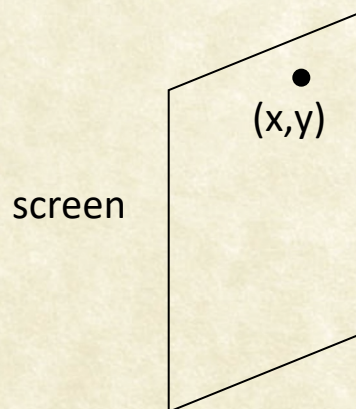
Known	Problem type
H, g	Recovery
g	Blind recovery
g, H partially	Semi blind recovery
f, g	System identification



Geometric Distortion



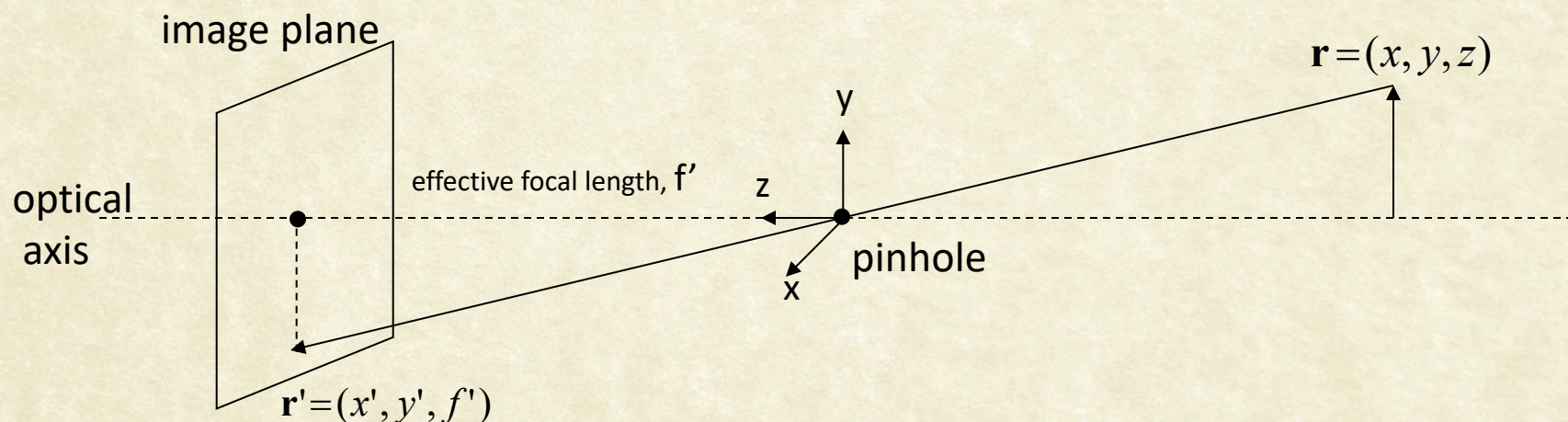
Pinhole and Perspective Projection



scene

Is an image being formed on the screen?

YES! But, not a “clear” one.

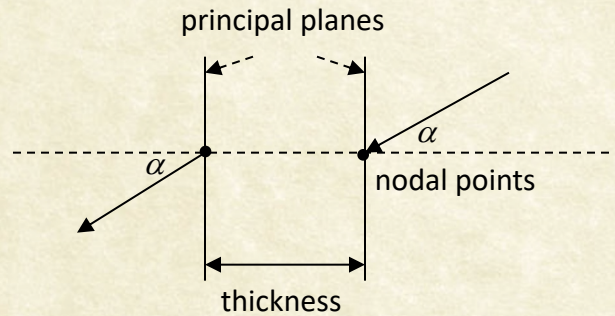


$$\frac{\mathbf{r}'}{f'} = \frac{\mathbf{r}}{z} \quad \Rightarrow \quad \frac{x'}{f'} = \frac{x}{z} \quad \frac{y'}{f'} = \frac{y}{z}$$

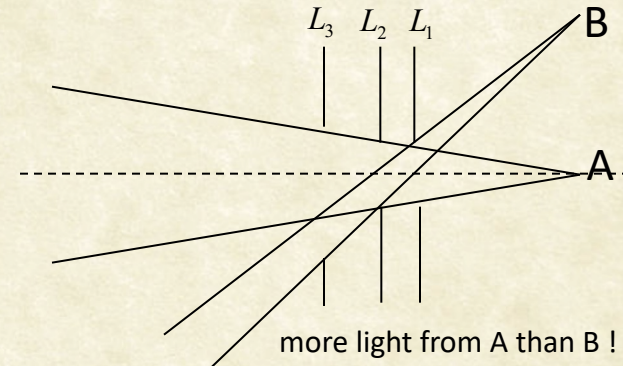


Common Lens Related Issues

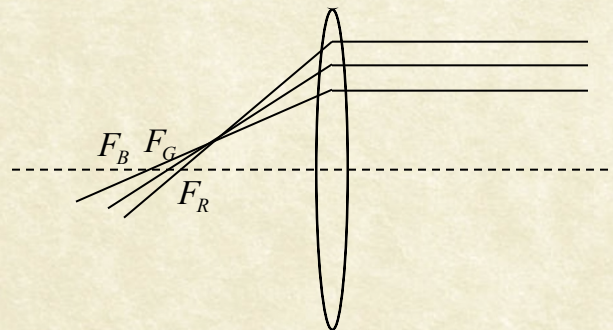
Compound (Thick) Lens



Vignetting

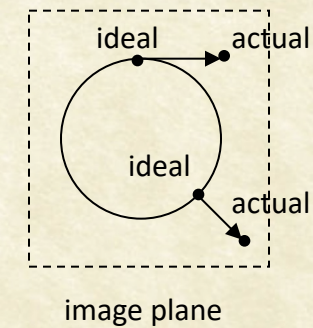


Chromatic Abberation



Lens has different refractive indices
for different wavelengths.

Radial and Tangential Distortion





Lens Glare



- Stray interreflections of light within the optical lens system.
- Happens when very bright sources are present in the scene.



Vignetting



photo by Robert Johnes



Chromatic Aberrations



longitudinal chromatic aberration
(axial)



transverse chromatic aberration
(lateral)



Geometric Lens Distortions

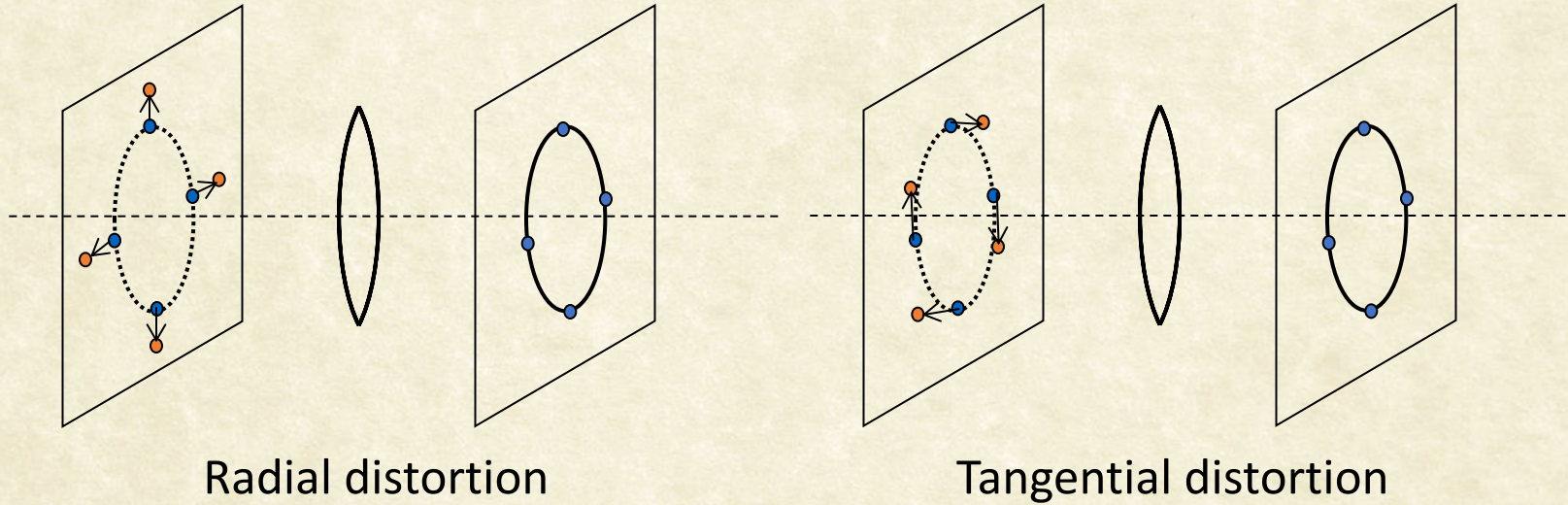


Photo by Helmut Dersch

Both due to lens imperfection
Rectify with geometric camera calibration



Is Distortion so bad?



Noise, Vignetting



Motion Blur



Barrel Distortion



Other Degradations





Motion Blur





An eye doctor's business card

M_____

HAS AN APPOINTMENT ON

☐ MON ☐ TUES ☐ WED ☐ THURS ☐ FRI ☐ SAT

DATE_____ATT_____A.M.
P.M.

IF UNABLE TO KEEP APPOINTMENT, KINDLY GIVE 24 HRS. NOTICE.



References

- <http://www.robots.ox.ac.uk/~az/lectures/ia/lect3.pdf>
- https://www.ece.iastate.edu/~namrata/EE528_Spring07/ImageRestoration1.pdf
- <http://www.ee.columbia.edu/~xli/ee4830/notes/lec7.pdf>
- DIP Ch. 5 (G&W)